CSE558 · DATA SCIENCE — LECTURE 10

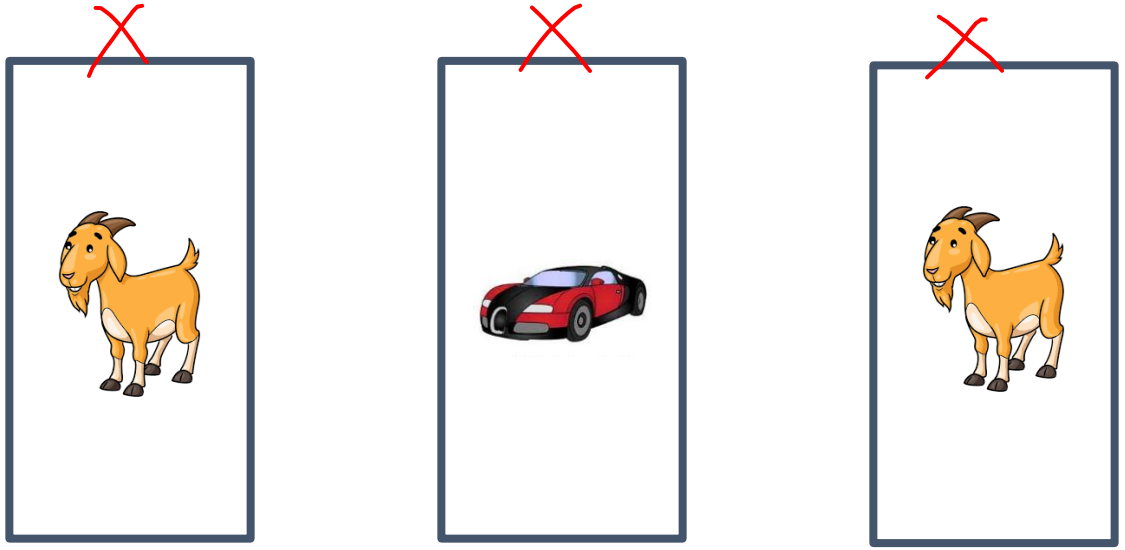
From Beliefs to Guarantees

Supratim · IIIT-Delhi · Monsoon 2026

THE CLASSIC CASE

Monty Hall

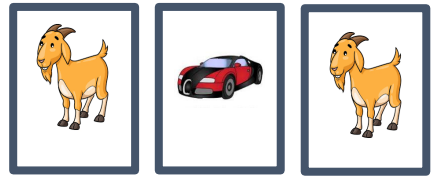
There is a car behind one of the three doors



Door 1 Door 2 Door 3

H: Door that has the car (H = 1, 2, 3).
E: Door that Monty reveals (E = 1, 2, 3)
Let Door1 is selected and Door3 revealed.
Compute, Prior, Likelihood, Marginal to get Posterior.

Do not change the option



H: Door that has the car (H = 1, 2, 3).
E: Door that Monty reveals (E = 1, 2, 3)
Let Door1 is selected and Door3 revealed.

Bayes rule

$$P_r(H=1 \mid E=3) = \frac{P_r(E=3 \mid H=1) \cdot P_r(H=1)}{P_r(E=3)}$$

$P_r(E=3)$

$P_r(E=3 \mid H=1) = 1/2$, $P_r(E=3 \mid H=2) = 1$, $P_r(E=3 \mid H=3) = 0$

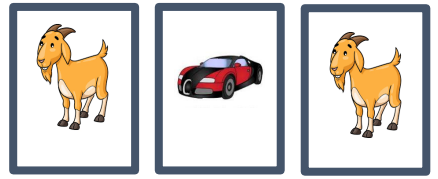
$P_r(H=1) = 1/3$

$$\sum_{i=1}^{3} P_r(E=3 \mid H=i)\, P_r(H=1) = \boxed{\frac{1}{2}\,\frac{1}{3}} + 1\,\frac{1}{3} + 0\,\frac{1}{3} = \frac{1}{2}$$

Likelihood Prior

$$= \frac{1/2 \; 1/3}{1/2} = 1/3$$

Changing the option



H: Door that has the car (H = 1, 2, 3).
E: Door that Monty reveals (E = 1, 2, 3)
Let Door1 is selected and Door3 revealed.

Change I choose door 2

$$P_v(E=3) = \sum_{i=1}^{3} P_v(E=3|H=1)\, P_v(H=1) = \frac{1}{2}$$

$$P_v(E=3|H=2) = 1$$

$$P_v(H=2|E=3) = \frac{P_v(E=3|H=2)\, P_0(H=2)}{P_v(E=3)} = \frac{1 \cdot 1/3}{1/2} = 2/3$$

Put a prior on the parameter itself

Lecture 9 — Bayes on a hypothesis H

$$\mathrm{P}(H \mid E) = \frac{\mathrm{P}(E \mid H)\,\mathrm{P}(H)}{\mathrm{P}(E)}$$

H: 'has the condition', 'car is behind Door 2'

Today — Bayes on a parameter θ

$$\mathrm{P}(\theta \mid \text{data}) = \frac{\mathrm{P}(\text{data} \mid \theta)\,\mathrm{P}(\theta)}{\mathrm{P}(\text{data})}$$

the coin's bias is p

The MAP estimate is the most probable θ once the data are in:

$$\hat{\theta}_{\mathrm{MAP}} = \arg\max_{\theta} \; \mathrm{P}(\text{data} \mid \theta) \cdot \mathrm{P}(\theta)$$

For comparison: $\hat{\theta}_{\mathrm{MLE}} = \arg\max_{\theta} \; \mathrm{P}(\text{data} \mid \theta)$

$$\arg\max_{\theta} P(\theta \mid \text{Data}) = \arg\max_{\theta} P(\text{Data} \mid \theta)\, P(\theta)$$

SAME COIN, NEW BELIEFS

Does a prior change the verdict?

Same coin, same data — 4 heads in 10 tosses. This time you walk in with a belief about the bias:

P(p = 1/3) = 0.10 **P(p = 1/2) = 0.20** **P(p = 2/3) = 0.70**

p	Likelihood P(4 heads | p)	Prior P(p)	MAP
1/3	0.228	0.10	0.028
1/2	0.205	0.20	0.041
2/3	0.057	0.70	0.039

$P_1(\theta) \propto P_0(\text{Data} \mid \theta)\, P_0(\theta)$

THE REVEAL

Neither the prior's pick nor the data's

Before any toss

p = 1/3	**0.10**
p = 1/2	**0.20**
p = 2/3	**0.70**

After 10 tosses · 4 heads

p = 1/3
p = 1/2
p = 2/3

p	Likelihood	Prior	Likelihood × Prior	Posterior
1/3	0.228	0.10	0.0228	
1/2	0.205	0.20	**0.0410**	
2/3	0.057	0.70	0.0399	

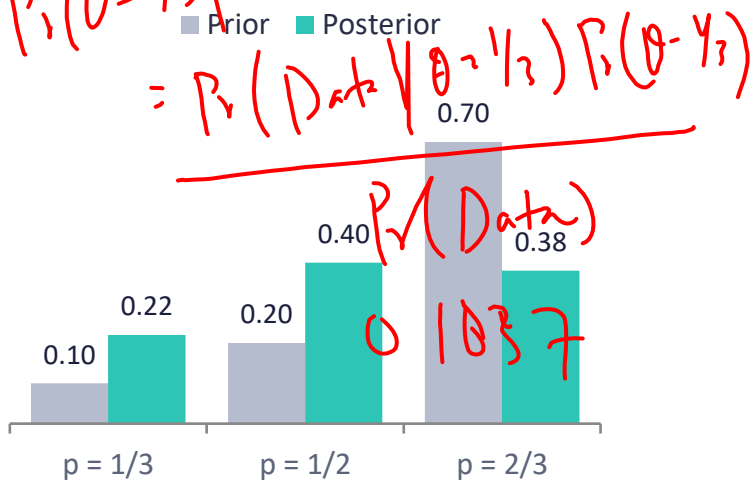
$P_r(\text{Data} \mid \theta)\, P(\theta)$ → New prior

0.1037

$P_r(\theta) \rightarrow P_r(\theta \mid \text{Data})$

↑ posterior

$P_r(\theta = 1/3 \mid \text{Data}) = \frac{P_r(\text{Data} \mid \theta = 1/3)\, P_r(\theta = 1/3)}{P_r(\text{Data})}$



0.1037

TODAY'S POSTERIOR IS TOMORROW'S PRIOR

Feed the same tosses in two batches

Before any toss

p = 1/3	**0.10**
p = 1/2	**0.20**
p = 2/3	**0.70**

After tosses 1–5 · 2 heads

p = 1/3
p = 1/2
p = 2/3

After tosses 6–10 · 2 more heads

p = 1/3
p = 1/2
p = 2/3

p	Likelihood	Prior	Likelihood × Prior	Posterior
1/3	0.330	0.10		
1/2	0.312	0.20		
2/3	0.165	0.70		

What a smooth prior does to the estimate

Now let the bias p be any value in [0, 1], with a prior density proportional to p(1 − p) :

$$\text{posterior}(p) \propto \underbrace{p^x(1-p)^{n-x}}_{\text{likelihood}} \times \underbrace{p(1-p)}_{\text{prior}} = p^{x+1}(1-p)^{n-x+1}$$

$$MAP(\theta) = \frac{p^x(1-p)^{n-x} \cdot p(1-p)}{} = p^{x+1}(1-p)^{n-x+1}$$

A PRIOR IS PSEUDO-DATA

One imaginary head, one imaginary tail

$$\hat{p}_{\mathrm{MAP}} = \frac{x+1}{n+2} \qquad \text{vs.} \qquad \hat{p}_{\mathrm{MLE}} = \frac{x}{n}$$

Data	MLE x / n	MAP (x + 1) / (n + 2)
0 heads in 3	0	0.20
4 heads in 10	0.40	0.42
400 heads in 1,000	0.4000	0.4002

The prior behaves exactly like two extra tosses you ‘saw’ before starting — one head and one tail.

SAME DATA, DIFFERENT QUESTION

MLE or MAP?

MLE — the data alone

- Easy to compute
- Consistent: θ → true θ as n grows
- Overfits small samples — 0 heads in 3 gives p̂ = 0
- The maximizer need not be unique

MAP — data plus a prior

- Regularized by the prior
- Resists overfitting small samples
- Converges to MLE as n → ∞ // if P(θ) > 0
- The answer depends on the prior you chose

Is the convergence always true?

$Pr(Data|\theta)$ θ_{mle} vs $Pr(Data|\theta)\,Pr(\theta)$ θ_{Map}

A NEW KIND OF QUESTION

From a best guess to a guarantee

Estimation handed us one number. Now: how far can a random quantity stray from its mean — with a guarantee that holds for every distribution, not just the Normal?

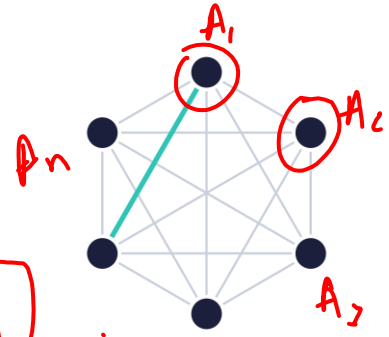
MORE INFORMATION ABOUT X → SHARPER BOUND

PUZZLE

Is the CAPTCHA vendor telling the truth?

1 **The claim**
The vendor's database holds 1,000,000 distinct CAPTCHA images.

2 **The mechanism**
Every security check shows one image, chosen uniformly at random.



$X_{ij} = \begin{cases} 1 & A_i = A_j \\ 0 & \text{else} \end{cases}$ $\quad (i \quad n)$ $\quad \binom{n}{2}$ $\quad 10^{-6}$

$E[\text{collisions}] = E\left[\sum_{i<j} X_{ij}\right]$

$= \binom{n}{2} 10^{-6}$

≈ 0.5

audit size

if $n = 1000$

$P_r(i = \text{captcha}) = \frac{1}{10^6}$

$P_r(j = \text{captcha}) = \frac{1}{10^6}$

$P(\text{Bot is, takes capt}) = 10^{-12}$

Only the mean — and still a guarantee

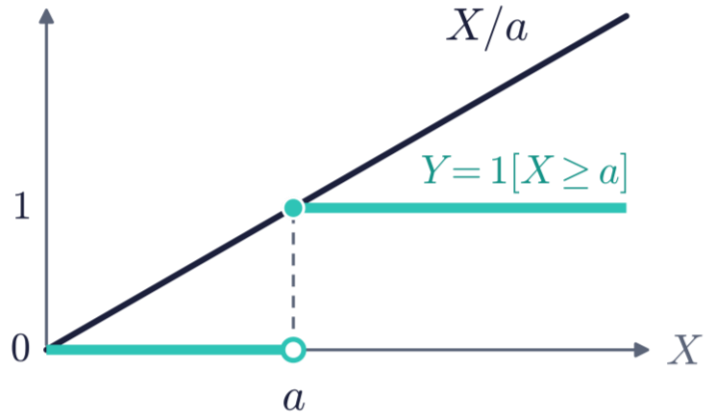
For any random variable X ≥ 0 and any a > 0: $P(X \geq a) \leq E[X]/a$

$$Y = \begin{cases} 1 & \text{if } X \geq a \\ 0 & \text{else} \end{cases}$$

① $\frac{X}{a} \geq Y$

② $E[Y] = 1 \cdot P_r(X \geq a) + 0 \cdot P_r(X < a)$

$$P_r(X \geq a) = E[Y] \leq E\left[\frac{X}{a}\right] \leq \frac{E[X]}{a}$$



A coin that lands heads 20% of the time

Toss it 100 times and let S be the number of heads. How likely are 50 or more heads?

$$X = \begin{cases} 1 & \text{w.p. } 0.2 \\ 0 & \text{else} \end{cases}$$

$$E[X] = 0.2$$

$$E[S] = 20$$

$$\Pr(S \geq 50) \leq \frac{E[S]}{50} = \frac{2}{5}$$

BACK TO THE PUZZLE

When should you stop believing the vendor?

With an audit of 1000 captcha

$$E[\#\text{collisions}] = 0.5$$

$$P_r(\#\text{collisions} \geq 10) \leq \frac{0.5}{10} = 0.05$$

Q To wrongly accuse an honest vendor at most 1% of the time, flag at how many repeats or more?

STUDY MATERIAL

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Probability and Computing: Chapter 3, by Mitzenmacher & Upfal

Slides